Exercise 2.6

1. Simplify the following using addition and subtraction properties of surds:
   (i) $5\sqrt{3}+18\sqrt{3}-2\sqrt{3}$ (ii) $4\sqrt[3]{5}+2\sqrt[3]{5}-3\sqrt[3]{5}$
   (iii) $3\sqrt{75}+5\sqrt{48}-\sqrt{243}$ (iv) $5\sqrt[3]{40}+2\sqrt[3]{625}-3\sqrt[3]{320}$

2. Simplify the following using multiplication and division properties of surds:
   (i) $\sqrt{3}\times\sqrt{5}\times\sqrt{2}$ (ii) $\sqrt{35}\div\sqrt{7}$ (iii) $\sqrt[3]{27}\times\sqrt[3]{8}\times\sqrt[3]{125}$
   (iv) $\left(7\sqrt{a}-5\sqrt{b}\right)\left(7\sqrt{a}+5\sqrt{b}\right)$ (v) $\left[\sqrt{\frac{225}{729}}-\sqrt{\frac{25}{144}}\right]\div\sqrt{\frac{16}{81}}$

3. If $\sqrt{2}=1.414,\ \sqrt{3}=1.732,\sqrt{5}=2.236,\sqrt{10}=3.162$, then find the values of the following correct to 3 places of decimals.
   (i) $\sqrt{40}-\sqrt{20}$ (ii) $\sqrt{300}+\sqrt{90}-\sqrt{8}$

4. Arrange surds in descending order : (i) $\sqrt[3]{5},\sqrt[9]{4},\sqrt[6]{3}$ (ii) $\sqrt[2]{\sqrt[3]{5}},\sqrt[3]{\sqrt[4]{7}},\sqrt{\sqrt{3}}$

5. Can you get a pure surd when you find
   (i) the sum of two surds (ii) the difference of two surds
   (iii) the product of two surds (iv) the quotient of two surds
   Justify each answer with an example.

6. Can you get a rational number when you compute
   (i) the sum of two surds (ii) the difference of two surds
   (iii) the product of two surds (iv) the quotient of two surds
   Justify each answer with an example.

2.7 Rationalisation of Surds

Rationalising factor is a term with which a term is multiplied or divided to make the whole term rational.

Examples:

(i) $\sqrt{3}$ is a rationalising factor of $\sqrt{3}$ (since $\sqrt{3}\times\sqrt{3}$ = the rational number 3)

(ii) $\sqrt[7]{5^4}$ is a rationalising factor of $\sqrt[7]{5^3}$ (since their product $=\sqrt[7]{5^7}=5$, a rational)

Thinking Corner

1. In the example (i) above, can $\sqrt{12}$ also be a rationalising factor? Can you think of any other number as a rationalising factor for $\sqrt{3}$?
2. Can you think of any other number as a rationalising factor for $\sqrt[7]{5^3}$ in example (ii) ?
3. If there can be many rationalising factors for an expression containing a surd, is there any advantage in choosing the smallest among them for manipulation?

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